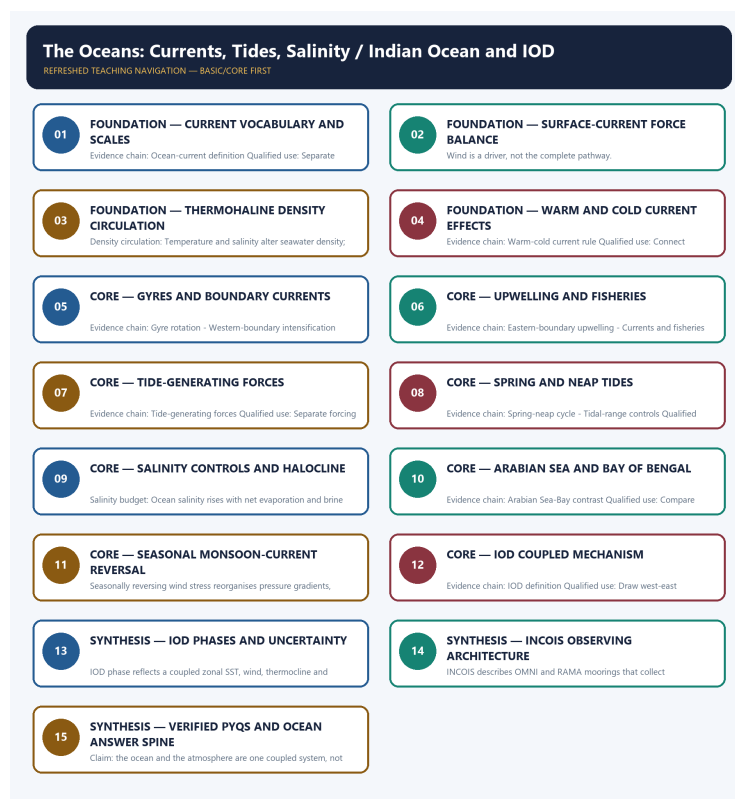


SOLVED PRACTICE WORKBOOK

The Oceans: Currents, Tides, Salinity / Indian Ocean and IOD - Learner-v2 Refreshed

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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BASIC MCQS / REMEDIATION

Q1. Which statement correctly explains Ocean-current definition?

A. An ocean current is a persistent, directed movement of seawater at a stated depth and time scale; a surface current, tidal stream, eddy and deep overturning branch are not interchangeable. B. Warm and cold describe a current relative to surrounding water and source region, not an absolute temperature; each current also changes air-sea heat and moisture exchange. C. Temperature and salinity alter seawater density; cooling, evaporation, sea-ice processes, mixing and freshwater input help create sinking, spreading and overturning at depth. D. Planetary winds provide major surface stress, while Coriolis, pressure gradients, basin shape, friction and sea-level slope organise the resulting circulation.

Answer: A. Explanation: An ocean current is a persistent, directed movement of seawater at a stated depth and time scale; a surface current, tidal stream, eddy and deep overturning branch are not interchangeable. The other options describe different processes, locations, scales or governance categories.

Q2. Which option is the safest spatial interpretation of Ocean-current definition?

A. Wind stress and Coriolis organise subtropical gyres clockwise in the Northern Hemisphere and anticlockwise in the Southern Hemisphere, with equatorial and seasonal complications. B. An ocean current is a persistent, directed movement of seawater at a stated depth and time scale; a surface current, tidal stream, eddy and deep overturning branch are not interchangeable. C. Warm and cold describe a current relative to surrounding water and source region, not an absolute temperature; each current also changes air-sea heat and moisture exchange. D. Temperature and salinity alter seawater density; cooling, evaporation, sea-ice processes, mixing and freshwater input help create sinking, spreading and overturning at depth.

Answer: B. Explanation: An ocean current is a persistent, directed movement of seawater at a stated depth and time scale; a surface current, tidal stream, eddy and deep overturning branch are not interchangeable. The other options describe different processes, locations, scales or governance categories.

Q3. Which statement preserves the process boundary for Ocean-current definition?

A. Warm and cold describe a current relative to surrounding water and source region, not an absolute temperature; each current also changes air-sea heat and moisture exchange. B. Wind stress and Coriolis organise subtropical gyres clockwise in the Northern Hemisphere and anticlockwise in the Southern Hemisphere, with equatorial and seasonal complications. C. An ocean current is a persistent, directed movement of seawater at a stated depth and time scale; a surface current, tidal stream, eddy and deep overturning branch are not interchangeable. D. Western boundary currents such as the Gulf Stream and Kuroshio are generally narrow, swift and deep because basin-scale vorticity balance concentrates return flow on the west.

Answer: C. Explanation: An ocean current is a persistent, directed movement of seawater at a stated depth and time scale; a surface current, tidal stream, eddy and deep overturning branch are not interchangeable. The other options describe different processes, locations, scales or governance categories.

Q4. Which option avoids the main UPSC trap concerning Ocean-current definition?

A. Eastern boundary currents are generally broad and cool, and alongshore winds can drive Ekman transport away from the coast so nutrient-rich deeper water rises. B. Western boundary currents such as the Gulf Stream and Kuroshio are generally narrow, swift and deep because basin-scale vorticity balance concentrates return flow on the west. C. Wind stress and Coriolis organise subtropical gyres clockwise in the Northern Hemisphere and anticlockwise in the Southern Hemisphere, with equatorial and seasonal complications. D. An ocean current is a persistent, directed movement of seawater at a stated depth and time scale; a surface current, tidal stream, eddy and deep overturning branch are not interchangeable.

Answer: D. Explanation: An ocean current is a persistent, directed movement of seawater at a stated depth and time scale; a surface current, tidal stream, eddy and deep overturning branch are not interchangeable. The other options describe different processes, locations, scales or governance categories.

Q5. Which statement correctly explains Surface-current drivers?

A. Planetary winds provide major surface stress, while Coriolis, pressure gradients, basin shape, friction and sea-level slope organise the resulting circulation. B. Temperature and salinity alter seawater density; cooling, evaporation, sea-ice processes, mixing and freshwater input help create sinking, spreading and overturning at depth. C. Wind stress and Coriolis organise subtropical gyres clockwise in the Northern Hemisphere and anticlockwise in the Southern Hemisphere, with equatorial and seasonal complications. D. Warm and cold describe a current relative to surrounding water and source region, not an absolute temperature; each current also changes air-sea heat and moisture exchange.

Answer: A. Explanation: Planetary winds provide major surface stress, while Coriolis, pressure gradients, basin shape, friction and sea-level slope organise the resulting circulation. The other options describe different processes, locations, scales or governance categories.

Q6. Which option is the safest spatial interpretation of Surface-current drivers?

A. Warm and cold describe a current relative to surrounding water and source region, not an absolute temperature; each current also changes air-sea heat and moisture exchange. B. Planetary winds provide major surface stress, while Coriolis, pressure gradients, basin shape, friction and sea-level slope organise the resulting circulation. C. Western boundary currents such as the Gulf Stream and Kuroshio are generally narrow, swift and deep because basin-scale vorticity balance concentrates return flow on the west. D. Wind stress and Coriolis organise subtropical gyres clockwise in the Northern Hemisphere and anticlockwise in the Southern Hemisphere, with equatorial and seasonal complications.

Answer: B. Explanation: Planetary winds provide major surface stress, while Coriolis, pressure gradients, basin shape, friction and sea-level slope organise the resulting circulation. The other options describe different processes, locations, scales or governance categories.

Q7. Which statement preserves the process boundary for Surface-current drivers?

A. Wind stress and Coriolis organise subtropical gyres clockwise in the Northern Hemisphere and anticlockwise in the Southern Hemisphere, with equatorial and seasonal complications. B. Western boundary currents such as the Gulf Stream and Kuroshio are generally narrow, swift and deep because basin-scale vorticity balance concentrates return flow on the west. C. Planetary winds provide major surface stress, while Coriolis, pressure gradients, basin shape, friction and sea-level slope organise the resulting circulation. D. Eastern boundary currents are generally broad and cool, and alongshore winds can drive Ekman transport away from the coast so nutrient-rich deeper water rises.

Answer: C. Explanation: Planetary winds provide major surface stress, while Coriolis, pressure gradients, basin shape, friction and sea-level slope organise the resulting circulation. The other options describe different processes, locations, scales or governance categories.

Q8. Which option avoids the main UPSC trap concerning Surface-current drivers?

A. Western boundary currents such as the Gulf Stream and Kuroshio are generally narrow, swift and deep because basin-scale vorticity balance concentrates return flow on the west. B. Productive fisheries often occur in upwelling zones or mixing fronts because nutrients support food webs; current convergence alone does not guarantee a rich fishery. C. Eastern boundary currents are generally broad and cool, and alongshore winds can drive Ekman transport away from the coast so nutrient-rich deeper water rises. D. Planetary winds provide major surface stress, while Coriolis, pressure gradients, basin shape, friction and sea-level slope organise the resulting circulation.

Answer: D. Explanation: Planetary winds provide major surface stress, while Coriolis, pressure gradients, basin shape, friction and sea-level slope organise the resulting circulation. The other options describe different processes, locations, scales or governance categories.

Q9. Which statement correctly explains Density circulation?

A. Temperature and salinity alter seawater density; cooling, evaporation, sea-ice processes, mixing and freshwater input help create sinking, spreading and overturning at depth. B. Western boundary currents such as the Gulf Stream and Kuroshio are generally narrow, swift and deep because basin-scale vorticity balance concentrates return flow on the west. C. Wind stress and Coriolis organise subtropical gyres clockwise in the Northern Hemisphere and anticlockwise in the Southern Hemisphere, with equatorial and seasonal complications. D. Warm and cold describe a current relative to surrounding water and source region, not an absolute temperature; each current also changes air-sea heat and moisture exchange.

Answer: A. Explanation: Temperature and salinity alter seawater density; cooling, evaporation, sea-ice processes, mixing and freshwater input help create sinking, spreading and overturning at depth. The other options describe different processes, locations, scales or governance categories.

Q10. Which option is the safest spatial interpretation of Density circulation?

A. Eastern boundary currents are generally broad and cool, and alongshore winds can drive Ekman transport away from the coast so nutrient-rich deeper water rises. B. Temperature and salinity alter seawater density; cooling, evaporation, sea-ice processes, mixing and freshwater input help create sinking, spreading and overturning at depth. C. Western boundary currents such as the Gulf Stream and Kuroshio are generally narrow, swift and deep because basin-scale vorticity balance concentrates return flow on the west. D. Wind stress and Coriolis organise subtropical gyres clockwise in the Northern Hemisphere and anticlockwise in the Southern Hemisphere, with equatorial and seasonal complications.

Answer: B. Explanation: Temperature and salinity alter seawater density; cooling, evaporation, sea-ice processes, mixing and freshwater input help create sinking, spreading and overturning at depth. The other options describe different processes, locations, scales or governance categories.

Q11. Which statement preserves the process boundary for Density circulation?

A. Productive fisheries often occur in upwelling zones or mixing fronts because nutrients support food webs; current convergence alone does not guarantee a rich fishery. B. Western boundary currents such as the Gulf Stream and Kuroshio are generally narrow, swift and deep because basin-scale vorticity balance concentrates return flow on the west. C. Temperature and salinity alter seawater density; cooling, evaporation, sea-ice processes, mixing and freshwater input help create sinking, spreading and overturning at depth. D. Eastern boundary currents are generally broad and cool, and alongshore winds can drive Ekman transport away from the coast so nutrient-rich deeper water rises.

Answer: C. Explanation: Temperature and salinity alter seawater density; cooling, evaporation, sea-ice processes, mixing and freshwater input help create sinking, spreading and overturning at depth. The other options describe different processes, locations, scales or governance categories.

Q12. Which option avoids the main UPSC trap concerning Density circulation?

A. Tides arise chiefly from the differential gravitational effects of the Moon and Sun together with Earth-Moon system dynamics, then are reshaped by basin geometry and friction. B. Productive fisheries often occur in upwelling zones or mixing fronts because nutrients support food webs; current convergence alone does not guarantee a rich fishery. C. Eastern boundary currents are generally broad and cool, and alongshore winds can drive Ekman transport away from the coast so nutrient-rich deeper water rises. D. Temperature and salinity alter seawater density; cooling, evaporation, sea-ice processes, mixing and freshwater input help create sinking, spreading and overturning at depth.

Answer: D. Explanation: Temperature and salinity alter seawater density; cooling, evaporation, sea-ice processes, mixing and freshwater input help create sinking, spreading and overturning at depth. The other options describe different processes, locations, scales or governance categories.

Q13. Which statement correctly explains Warm-cold current rule?

A. Warm and cold describe a current relative to surrounding water and source region, not an absolute temperature; each current also changes air-sea heat and moisture exchange. B. Western boundary currents such as the Gulf Stream and Kuroshio are generally narrow, swift and deep because basin-scale vorticity balance concentrates return flow on the west. C. Wind stress and Coriolis organise subtropical gyres clockwise in the Northern Hemisphere and anticlockwise in the Southern Hemisphere, with equatorial and seasonal complications. D. Eastern boundary currents are generally broad and cool, and alongshore winds can drive Ekman transport away from the coast so nutrient-rich deeper water rises.

Answer: A. Explanation: Warm and cold describe a current relative to surrounding water and source region, not an absolute temperature; each current also changes air-sea heat and moisture exchange. The other options describe different processes, locations, scales or governance categories.

Q14. Which option is the safest spatial interpretation of Warm-cold current rule?

A. Western boundary currents such as the Gulf Stream and Kuroshio are generally narrow, swift and deep because basin-scale vorticity balance concentrates return flow on the west. B. Warm and cold describe a current relative to surrounding water and source region, not an absolute temperature; each current also changes air-sea heat and moisture exchange. C. Productive fisheries often occur in upwelling zones or mixing fronts because nutrients support food webs; current convergence alone does not guarantee a rich fishery. D. Eastern boundary currents are generally broad and cool, and alongshore winds can drive Ekman transport away from the coast so nutrient-rich deeper water rises.

Answer: B. Explanation: Warm and cold describe a current relative to surrounding water and source region, not an absolute temperature; each current also changes air-sea heat and moisture exchange. The other options describe different processes, locations, scales or governance categories.

Q15. Which statement preserves the process boundary for Warm-cold current rule?

A. Productive fisheries often occur in upwelling zones or mixing fronts because nutrients support food webs; current convergence alone does not guarantee a rich fishery. B. Tides arise chiefly from the differential gravitational effects of the Moon and Sun together with Earth-Moon system dynamics, then are reshaped by basin geometry and friction. C. Warm and cold describe a current relative to surrounding water and source region, not an absolute temperature; each current also changes air-sea heat and moisture exchange. D. Eastern boundary currents are generally broad and cool, and alongshore winds can drive Ekman transport away from the coast so nutrient-rich deeper water rises.

Answer: C. Explanation: Warm and cold describe a current relative to surrounding water and source region, not an absolute temperature; each current also changes air-sea heat and moisture exchange. The other options describe different processes, locations, scales or governance categories.

Q16. Which option avoids the main UPSC trap concerning Warm-cold current rule?

A. Tides arise chiefly from the differential gravitational effects of the Moon and Sun together with Earth-Moon system dynamics, then are reshaped by basin geometry and friction. B. Productive fisheries often occur in upwelling zones or mixing fronts because nutrients support food webs; current convergence alone does not guarantee a rich fishery. C. Spring tides occur near new and full moon when solar and lunar tidal effects reinforce; neap tides occur near quarter moons when the effects partly offset. D. Warm and cold describe a current relative to surrounding water and source region, not an absolute temperature; each current also changes air-sea heat and moisture exchange.

Answer: D. Explanation: Warm and cold describe a current relative to surrounding water and source region, not an absolute temperature; each current also changes air-sea heat and moisture exchange. The other options describe different processes, locations, scales or governance categories.

Q17. Which statement correctly explains Gyre rotation?

A. Wind stress and Coriolis organise subtropical gyres clockwise in the Northern Hemisphere and anticlockwise in the Southern Hemisphere, with equatorial and seasonal complications. B. Productive fisheries often occur in upwelling zones or mixing fronts because nutrients support food webs; current convergence alone does not guarantee a rich fishery. C. Western boundary currents such as the Gulf Stream and Kuroshio are generally narrow, swift and deep because basin-scale vorticity balance concentrates return flow on the west. D. Eastern boundary currents are generally broad and cool, and alongshore winds can drive Ekman transport away from the coast so nutrient-rich deeper water rises.

Answer: A. Explanation: Wind stress and Coriolis organise subtropical gyres clockwise in the Northern Hemisphere and anticlockwise in the Southern Hemisphere, with equatorial and seasonal complications. The other options describe different processes, locations, scales or governance categories.

Q18. Which option is the safest spatial interpretation of Gyre rotation?

A. Tides arise chiefly from the differential gravitational effects of the Moon and Sun together with Earth-Moon system dynamics, then are reshaped by basin geometry and friction. B. Wind stress and Coriolis organise subtropical gyres clockwise in the Northern Hemisphere and anticlockwise in the Southern Hemisphere, with equatorial and seasonal complications. C. Eastern boundary currents are generally broad and cool, and alongshore winds can drive Ekman transport away from the coast so nutrient-rich deeper water rises. D. Productive fisheries often occur in upwelling zones or mixing fronts because nutrients support food webs; current convergence alone does not guarantee a rich fishery.

Answer: B. Explanation: Wind stress and Coriolis organise subtropical gyres clockwise in the Northern Hemisphere and anticlockwise in the Southern Hemisphere, with equatorial and seasonal complications. The other options describe different processes, locations, scales or governance categories.

Q19. Which statement preserves the process boundary for Gyre rotation?

A. Productive fisheries often occur in upwelling zones or mixing fronts because nutrients support food webs; current convergence alone does not guarantee a rich fishery. B. Spring tides occur near new and full moon when solar and lunar tidal effects reinforce; neap tides occur near quarter moons when the effects partly offset. C. Wind stress and Coriolis organise subtropical gyres clockwise in the Northern Hemisphere and anticlockwise in the Southern Hemisphere, with equatorial and seasonal complications. D. Tides arise chiefly from the differential gravitational effects of the Moon and Sun together with Earth-Moon system dynamics, then are reshaped by basin geometry and friction.

Answer: C. Explanation: Wind stress and Coriolis organise subtropical gyres clockwise in the Northern Hemisphere and anticlockwise in the Southern Hemisphere, with equatorial and seasonal complications. The other options describe different processes, locations, scales or governance categories.

Q20. Which option avoids the main UPSC trap concerning Gyre rotation?

A. Tidal range depends on astronomical forcing, coastline and basin resonance, shelf depth, friction and local timing; funnel-shaped bays can amplify range. B. Spring tides occur near new and full moon when solar and lunar tidal effects reinforce; neap tides occur near quarter moons when the effects partly offset. C. Tides arise chiefly from the differential gravitational effects of the Moon and Sun together with Earth-Moon system dynamics, then are reshaped by basin geometry and friction. D. Wind stress and Coriolis organise subtropical gyres clockwise in the Northern Hemisphere and anticlockwise in the Southern Hemisphere, with equatorial and seasonal complications.

Answer: D. Explanation: Wind stress and Coriolis organise subtropical gyres clockwise in the Northern Hemisphere and anticlockwise in the Southern Hemisphere, with equatorial and seasonal complications. The other options describe different processes, locations, scales or governance categories.

Q21. Which statement correctly explains Western-boundary intensification?

A. Western boundary currents such as the Gulf Stream and Kuroshio are generally narrow, swift and deep because basin-scale vorticity balance concentrates return flow on the west. B. Tides arise chiefly from the differential gravitational effects of the Moon and Sun together with Earth-Moon system dynamics, then are reshaped by basin geometry and friction. C. Productive fisheries often occur in upwelling zones or mixing fronts because nutrients support food webs; current convergence alone does not guarantee a rich fishery. D. Eastern boundary currents are generally broad and cool, and alongshore winds can drive Ekman transport away from the coast so nutrient-rich deeper water rises.

Answer: A. Explanation: Western boundary currents such as the Gulf Stream and Kuroshio are generally narrow, swift and deep because basin-scale vorticity balance concentrates return flow on the west. The other options describe different processes, locations, scales or governance categories.

Q22. Which option is the safest spatial interpretation of Western-boundary intensification?

A. Spring tides occur near new and full moon when solar and lunar tidal effects reinforce; neap tides occur near quarter moons when the effects partly offset. B. Western boundary currents such as the Gulf Stream and Kuroshio are generally narrow, swift and deep because basin-scale vorticity balance concentrates return flow on the west. C. Productive fisheries often occur in upwelling zones or mixing fronts because nutrients support food webs; current convergence alone does not guarantee a rich fishery. D. Tides arise chiefly from the differential gravitational effects of the Moon and Sun together with Earth-Moon system dynamics, then are reshaped by basin geometry and friction.

Answer: B. Explanation: Western boundary currents such as the Gulf Stream and Kuroshio are generally narrow, swift and deep because basin-scale vorticity balance concentrates return flow on the west. The other options describe different processes, locations, scales or governance categories.

Q23. Which statement preserves the process boundary for Western-boundary intensification?

A. Tides arise chiefly from the differential gravitational effects of the Moon and Sun together with Earth-Moon system dynamics, then are reshaped by basin geometry and friction. B. Spring tides occur near new and full moon when solar and lunar tidal effects reinforce; neap tides occur near quarter moons when the effects partly offset. C. Western boundary currents such as the Gulf Stream and Kuroshio are generally narrow, swift and deep because basin-scale vorticity balance concentrates return flow on the west. D. Tidal range depends on astronomical forcing, coastline and basin resonance, shelf depth, friction and local timing; funnel-shaped bays can amplify range.

Answer: C. Explanation: Western boundary currents such as the Gulf Stream and Kuroshio are generally narrow, swift and deep because basin-scale vorticity balance concentrates return flow on the west. The other options describe different processes, locations, scales or governance categories.

Q24. Which option avoids the main UPSC trap concerning Western-boundary intensification?

A. Spring tides occur near new and full moon when solar and lunar tidal effects reinforce; neap tides occur near quarter moons when the effects partly offset. B. Ocean salinity rises with net evaporation and brine formation and falls with precipitation, river runoff, ice melt and mixing; it is a conservative tracer only within stated processes. C. Tidal range depends on astronomical forcing, coastline and basin resonance, shelf depth, friction and local timing; funnel-shaped bays can amplify range. D. Western boundary currents such as the Gulf Stream and Kuroshio are generally narrow, swift and deep because basin-scale vorticity balance concentrates return flow on the west.

Answer: D. Explanation: Western boundary currents such as the Gulf Stream and Kuroshio are generally narrow, swift and deep because basin-scale vorticity balance concentrates return flow on the west. The other options describe different processes, locations, scales or governance categories.

Q25. Which statement correctly explains Eastern-boundary upwelling?

A. Eastern boundary currents are generally broad and cool, and alongshore winds can drive Ekman transport away from the coast so nutrient-rich deeper water rises. B. Productive fisheries often occur in upwelling zones or mixing fronts because nutrients support food webs; current convergence alone does not guarantee a rich fishery. C. Spring tides occur near new and full moon when solar and lunar tidal effects reinforce; neap tides occur near quarter moons when the effects partly offset. D. Tides arise chiefly from the differential gravitational effects of the Moon and Sun together with Earth-Moon system dynamics, then are reshaped by basin geometry and friction.

Answer: A. Explanation: Eastern boundary currents are generally broad and cool, and alongshore winds can drive Ekman transport away from the coast so nutrient-rich deeper water rises. The other options describe different processes, locations, scales or governance categories.

Q26. Which option is the safest spatial interpretation of Eastern-boundary upwelling?

A. Spring tides occur near new and full moon when solar and lunar tidal effects reinforce; neap tides occur near quarter moons when the effects partly offset. B. Eastern boundary currents are generally broad and cool, and alongshore winds can drive Ekman transport away from the coast so nutrient-rich deeper water rises. C. Tidal range depends on astronomical forcing, coastline and basin resonance, shelf depth, friction and local timing; funnel-shaped bays can amplify range. D. Tides arise chiefly from the differential gravitational effects of the Moon and Sun together with Earth-Moon system dynamics, then are reshaped by basin geometry and friction.

Answer: B. Explanation: Eastern boundary currents are generally broad and cool, and alongshore winds can drive Ekman transport away from the coast so nutrient-rich deeper water rises. The other options describe different processes, locations, scales or governance categories.

Q27. Which statement preserves the process boundary for Eastern-boundary upwelling?

A. Tidal range depends on astronomical forcing, coastline and basin resonance, shelf depth, friction and local timing; funnel-shaped bays can amplify range. B. Ocean salinity rises with net evaporation and brine formation and falls with precipitation, river runoff, ice melt and mixing; it is a conservative tracer only within stated processes. C. Eastern boundary currents are generally broad and cool, and alongshore winds can drive Ekman transport away from the coast so nutrient-rich deeper water rises. D. Spring tides occur near new and full moon when solar and lunar tidal effects reinforce; neap tides occur near quarter moons when the effects partly offset.

Answer: C. Explanation: Eastern boundary currents are generally broad and cool, and alongshore winds can drive Ekman transport away from the coast so nutrient-rich deeper water rises. The other options describe different processes, locations, scales or governance categories.

Q28. Which option avoids the main UPSC trap concerning Eastern-boundary upwelling?

A. A halocline is a strong vertical salinity gradient; together with temperature it contributes to stratification, mixing resistance and the depth of the mixed layer. B. Tidal range depends on astronomical forcing, coastline and basin resonance, shelf depth, friction and local timing; funnel-shaped bays can amplify range. C. Ocean salinity rises with net evaporation and brine formation and falls with precipitation, river runoff, ice melt and mixing; it is a conservative tracer only within stated processes. D. Eastern boundary currents are generally broad and cool, and alongshore winds can drive Ekman transport away from the coast so nutrient-rich deeper water rises.

Answer: D. Explanation: Eastern boundary currents are generally broad and cool, and alongshore winds can drive Ekman transport away from the coast so nutrient-rich deeper water rises. The other options describe different processes, locations, scales or governance categories.

Q29. Which statement correctly explains Currents and fisheries?

A. Productive fisheries often occur in upwelling zones or mixing fronts because nutrients support food webs; current convergence alone does not guarantee a rich fishery. B. Tides arise chiefly from the differential gravitational effects of the Moon and Sun together with Earth-Moon system dynamics, then are reshaped by basin geometry and friction. C. Tidal range depends on astronomical forcing, coastline and basin resonance, shelf depth, friction and local timing; funnel-shaped bays can amplify range. D. Spring tides occur near new and full moon when solar and lunar tidal effects reinforce; neap tides occur near quarter moons when the effects partly offset.

Answer: A. Explanation: Productive fisheries often occur in upwelling zones or mixing fronts because nutrients support food webs; current convergence alone does not guarantee a rich fishery. The other options describe different processes, locations, scales or governance categories.

Q30. Which option is the safest spatial interpretation of Currents and fisheries?

A. Tidal range depends on astronomical forcing, coastline and basin resonance, shelf depth, friction and local timing; funnel-shaped bays can amplify range. B. Productive fisheries often occur in upwelling zones or mixing fronts because nutrients support food webs; current convergence alone does not guarantee a rich fishery. C. Spring tides occur near new and full moon when solar and lunar tidal effects reinforce; neap tides occur near quarter moons when the effects partly offset. D. Ocean salinity rises with net evaporation and brine formation and falls with precipitation, river runoff, ice melt and mixing; it is a conservative tracer only within stated processes.

Answer: B. Explanation: Productive fisheries often occur in upwelling zones or mixing fronts because nutrients support food webs; current convergence alone does not guarantee a rich fishery. The other options describe different processes, locations, scales or governance categories.

Q31. Which statement preserves the process boundary for Currents and fisheries?

A. A halocline is a strong vertical salinity gradient; together with temperature it contributes to stratification, mixing resistance and the depth of the mixed layer. B. Ocean salinity rises with net evaporation and brine formation and falls with precipitation, river runoff, ice melt and mixing; it is a conservative tracer only within stated processes. C. Productive fisheries often occur in upwelling zones or mixing fronts because nutrients support food webs; current convergence alone does not guarantee a rich fishery. D. Tidal range depends on astronomical forcing, coastline and basin resonance, shelf depth, friction and local timing; funnel-shaped bays can amplify range.

Answer: C. Explanation: Productive fisheries often occur in upwelling zones or mixing fronts because nutrients support food webs; current convergence alone does not guarantee a rich fishery. The other options describe different processes, locations, scales or governance categories.

Q32. Which option avoids the main UPSC trap concerning Currents and fisheries?

A. A halocline is a strong vertical salinity gradient; together with temperature it contributes to stratification, mixing resistance and the depth of the mixed layer. B. The Arabian Sea is generally more saline than the Bay of Bengal because evaporation is stronger and freshwater input lower, while the Bay receives heavy rain and large river discharge. C. Ocean salinity rises with net evaporation and brine formation and falls with precipitation, river runoff, ice melt and mixing; it is a conservative tracer only within stated processes. D. Productive fisheries often occur in upwelling zones or mixing fronts because nutrients support food webs; current convergence alone does not guarantee a rich fishery.

Answer: D. Explanation: Productive fisheries often occur in upwelling zones or mixing fronts because nutrients support food webs; current convergence alone does not guarantee a rich fishery. The other options describe different processes, locations, scales or governance categories.

Q33. Which statement correctly explains Tide-generating forces?

A. Tides arise chiefly from the differential gravitational effects of the Moon and Sun together with Earth-Moon system dynamics, then are reshaped by basin geometry and friction. B. Spring tides occur near new and full moon when solar and lunar tidal effects reinforce; neap tides occur near quarter moons when the effects partly offset. C. Ocean salinity rises with net evaporation and brine formation and falls with precipitation, river runoff, ice melt and mixing; it is a conservative tracer only within stated processes. D. Tidal range depends on astronomical forcing, coastline and basin resonance, shelf depth, friction and local timing; funnel-shaped bays can amplify range.

Answer: A. Explanation: Tides arise chiefly from the differential gravitational effects of the Moon and Sun together with Earth-Moon system dynamics, then are reshaped by basin geometry and friction. The other options describe different processes, locations, scales or governance categories.

Q34. Which option is the safest spatial interpretation of Tide-generating forces?

A. Tidal range depends on astronomical forcing, coastline and basin resonance, shelf depth, friction and local timing; funnel-shaped bays can amplify range. B. Tides arise chiefly from the differential gravitational effects of the Moon and Sun together with Earth-Moon system dynamics, then are reshaped by basin geometry and friction. C. A halocline is a strong vertical salinity gradient; together with temperature it contributes to stratification, mixing resistance and the depth of the mixed layer. D. Ocean salinity rises with net evaporation and brine formation and falls with precipitation, river runoff, ice melt and mixing; it is a conservative tracer only within stated processes.

Answer: B. Explanation: Tides arise chiefly from the differential gravitational effects of the Moon and Sun together with Earth-Moon system dynamics, then are reshaped by basin geometry and friction. The other options describe different processes, locations, scales or governance categories.

Q35. Which statement preserves the process boundary for Tide-generating forces?

A. A halocline is a strong vertical salinity gradient; together with temperature it contributes to stratification, mixing resistance and the depth of the mixed layer. B. Ocean salinity rises with net evaporation and brine formation and falls with precipitation, river runoff, ice melt and mixing; it is a conservative tracer only within stated processes. C. Tides arise chiefly from the differential gravitational effects of the Moon and Sun together with Earth-Moon system dynamics, then are reshaped by basin geometry and friction. D. The Arabian Sea is generally more saline than the Bay of Bengal because evaporation is stronger and freshwater input lower, while the Bay receives heavy rain and large river discharge.

Answer: C. Explanation: Tides arise chiefly from the differential gravitational effects of the Moon and Sun together with Earth-Moon system dynamics, then are reshaped by basin geometry and friction. The other options describe different processes, locations, scales or governance categories.

Q36. Which option avoids the main UPSC trap concerning Tide-generating forces?

A. North Indian Ocean surface circulation reverses or reorganises seasonally with monsoon winds, unlike a single fixed subtropical-gyre arrow map. B. The Arabian Sea is generally more saline than the Bay of Bengal because evaporation is stronger and freshwater input lower, while the Bay receives heavy rain and large river discharge. C. A halocline is a strong vertical salinity gradient; together with temperature it contributes to stratification, mixing resistance and the depth of the mixed layer. D. Tides arise chiefly from the differential gravitational effects of the Moon and Sun together with Earth-Moon system dynamics, then are reshaped by basin geometry and friction.

Answer: D. Explanation: Tides arise chiefly from the differential gravitational effects of the Moon and Sun together with Earth-Moon system dynamics, then are reshaped by basin geometry and friction. The other options describe different processes, locations, scales or governance categories.

Q37. Which statement correctly explains Spring-neap cycle?

A. Spring tides occur near new and full moon when solar and lunar tidal effects reinforce; neap tides occur near quarter moons when the effects partly offset. B. Ocean salinity rises with net evaporation and brine formation and falls with precipitation, river runoff, ice melt and mixing; it is a conservative tracer only within stated processes. C. A halocline is a strong vertical salinity gradient; together with temperature it contributes to stratification, mixing resistance and the depth of the mixed layer. D. Tidal range depends on astronomical forcing, coastline and basin resonance, shelf depth, friction and local timing; funnel-shaped bays can amplify range.

Answer: A. Explanation: Spring tides occur near new and full moon when solar and lunar tidal effects reinforce; neap tides occur near quarter moons when the effects partly offset. The other options describe different processes, locations, scales or governance categories.

Q38. Which option is the safest spatial interpretation of Spring-neap cycle?

A. Ocean salinity rises with net evaporation and brine formation and falls with precipitation, river runoff, ice melt and mixing; it is a conservative tracer only within stated processes. B. Spring tides occur near new and full moon when solar and lunar tidal effects reinforce; neap tides occur near quarter moons when the effects partly offset. C. A halocline is a strong vertical salinity gradient; together with temperature it contributes to stratification, mixing resistance and the depth of the mixed layer. D. The Arabian Sea is generally more saline than the Bay of Bengal because evaporation is stronger and freshwater input lower, while the Bay receives heavy rain and large river discharge.

Answer: B. Explanation: Spring tides occur near new and full moon when solar and lunar tidal effects reinforce; neap tides occur near quarter moons when the effects partly offset. The other options describe different processes, locations, scales or governance categories.

Q39. Which statement preserves the process boundary for Spring-neap cycle?

A. A halocline is a strong vertical salinity gradient; together with temperature it contributes to stratification, mixing resistance and the depth of the mixed layer. B. North Indian Ocean surface circulation reverses or reorganises seasonally with monsoon winds, unlike a single fixed subtropical-gyre arrow map. C. Spring tides occur near new and full moon when solar and lunar tidal effects reinforce; neap tides occur near quarter moons when the effects partly offset. D. The Arabian Sea is generally more saline than the Bay of Bengal because evaporation is stronger and freshwater input lower, while the Bay receives heavy rain and large river discharge.

Answer: C. Explanation: Spring tides occur near new and full moon when solar and lunar tidal effects reinforce; neap tides occur near quarter moons when the effects partly offset. The other options describe different processes, locations, scales or governance categories.

Q40. Which option avoids the main UPSC trap concerning Spring-neap cycle?

A. The Arabian Sea is generally more saline than the Bay of Bengal because evaporation is stronger and freshwater input lower, while the Bay receives heavy rain and large river discharge. B. During the southwest monsoon, strong winds and the Somali Current support coastal upwelling in the western Arabian Sea, linking circulation to nutrients and air-sea exchange. C. North Indian Ocean surface circulation reverses or reorganises seasonally with monsoon winds, unlike a single fixed subtropical-gyre arrow map. D. Spring tides occur near new and full moon when solar and lunar tidal effects reinforce; neap tides occur near quarter moons when the effects partly offset.

Answer: D. Explanation: Spring tides occur near new and full moon when solar and lunar tidal effects reinforce; neap tides occur near quarter moons when the effects partly offset. The other options describe different processes, locations, scales or governance categories.

Q41. Which statement correctly explains Tidal-range controls?

A. Tidal range depends on astronomical forcing, coastline and basin resonance, shelf depth, friction and local timing; funnel-shaped bays can amplify range. B. A halocline is a strong vertical salinity gradient; together with temperature it contributes to stratification, mixing resistance and the depth of the mixed layer. C. The Arabian Sea is generally more saline than the Bay of Bengal because evaporation is stronger and freshwater input lower, while the Bay receives heavy rain and large river discharge. D. Ocean salinity rises with net evaporation and brine formation and falls with precipitation, river runoff, ice melt and mixing; it is a conservative tracer only within stated processes.

Answer: A. Explanation: Tidal range depends on astronomical forcing, coastline and basin resonance, shelf depth, friction and local timing; funnel-shaped bays can amplify range. The other options describe different processes, locations, scales or governance categories.

Q42. Which option is the safest spatial interpretation of Tidal-range controls?

A. North Indian Ocean surface circulation reverses or reorganises seasonally with monsoon winds, unlike a single fixed subtropical-gyre arrow map. B. Tidal range depends on astronomical forcing, coastline and basin resonance, shelf depth, friction and local timing; funnel-shaped bays can amplify range. C. The Arabian Sea is generally more saline than the Bay of Bengal because evaporation is stronger and freshwater input lower, while the Bay receives heavy rain and large river discharge. D. A halocline is a strong vertical salinity gradient; together with temperature it contributes to stratification, mixing resistance and the depth of the mixed layer.

Answer: B. Explanation: Tidal range depends on astronomical forcing, coastline and basin resonance, shelf depth, friction and local timing; funnel-shaped bays can amplify range. The other options describe different processes, locations, scales or governance categories.

Q43. Which statement preserves the process boundary for Tidal-range controls?

A. The Arabian Sea is generally more saline than the Bay of Bengal because evaporation is stronger and freshwater input lower, while the Bay receives heavy rain and large river discharge. B. During the southwest monsoon, strong winds and the Somali Current support coastal upwelling in the western Arabian Sea, linking circulation to nutrients and air-sea exchange. C. Tidal range depends on astronomical forcing, coastline and basin resonance, shelf depth, friction and local timing; funnel-shaped bays can amplify range. D. North Indian Ocean surface circulation reverses or reorganises seasonally with monsoon winds, unlike a single fixed subtropical-gyre arrow map.

Answer: C. Explanation: Tidal range depends on astronomical forcing, coastline and basin resonance, shelf depth, friction and local timing; funnel-shaped bays can amplify range. The other options describe different processes, locations, scales or governance categories.

Q44. Which option avoids the main UPSC trap concerning Tidal-range controls?

A. During the southwest monsoon, strong winds and the Somali Current support coastal upwelling in the western Arabian Sea, linking circulation to nutrients and air-sea exchange. B. The Indian Ocean Dipole is a coupled zonal contrast in sea-surface temperature and associated atmosphere-ocean conditions between western and eastern equatorial Indian Ocean sectors. C. North Indian Ocean surface circulation reverses or reorganises seasonally with monsoon winds, unlike a single fixed subtropical-gyre arrow map. D. Tidal range depends on astronomical forcing, coastline and basin resonance, shelf depth, friction and local timing; funnel-shaped bays can amplify range.

Answer: D. Explanation: Tidal range depends on astronomical forcing, coastline and basin resonance, shelf depth, friction and local timing; funnel-shaped bays can amplify range. The other options describe different processes, locations, scales or governance categories.

Q45. Which statement correctly explains Salinity budget?

A. Ocean salinity rises with net evaporation and brine formation and falls with precipitation, river runoff, ice melt and mixing; it is a conservative tracer only within stated processes. B. North Indian Ocean surface circulation reverses or reorganises seasonally with monsoon winds, unlike a single fixed subtropical-gyre arrow map. C. A halocline is a strong vertical salinity gradient; together with temperature it contributes to stratification, mixing resistance and the depth of the mixed layer. D. The Arabian Sea is generally more saline than the Bay of Bengal because evaporation is stronger and freshwater input lower, while the Bay receives heavy rain and large river discharge.

Answer: A. Explanation: Ocean salinity rises with net evaporation and brine formation and falls with precipitation, river runoff, ice melt and mixing; it is a conservative tracer only within stated processes. The other options describe different processes, locations, scales or governance categories.

Q46. Which option is the safest spatial interpretation of Salinity budget?

A. During the southwest monsoon, strong winds and the Somali Current support coastal upwelling in the western Arabian Sea, linking circulation to nutrients and air-sea exchange. B. Ocean salinity rises with net evaporation and brine formation and falls with precipitation, river runoff, ice melt and mixing; it is a conservative tracer only within stated processes. C. North Indian Ocean surface circulation reverses or reorganises seasonally with monsoon winds, unlike a single fixed subtropical-gyre arrow map. D. The Arabian Sea is generally more saline than the Bay of Bengal because evaporation is stronger and freshwater input lower, while the Bay receives heavy rain and large river discharge.

Answer: B. Explanation: Ocean salinity rises with net evaporation and brine formation and falls with precipitation, river runoff, ice melt and mixing; it is a conservative tracer only within stated processes. The other options describe different processes, locations, scales or governance categories.

Q47. Which statement preserves the process boundary for Salinity budget?

A. During the southwest monsoon, strong winds and the Somali Current support coastal upwelling in the western Arabian Sea, linking circulation to nutrients and air-sea exchange. B. The Indian Ocean Dipole is a coupled zonal contrast in sea-surface temperature and associated atmosphere-ocean conditions between western and eastern equatorial Indian Ocean sectors. C. Ocean salinity rises with net evaporation and brine formation and falls with precipitation, river runoff, ice melt and mixing; it is a conservative tracer only within stated processes. D. North Indian Ocean surface circulation reverses or reorganises seasonally with monsoon winds, unlike a single fixed subtropical-gyre arrow map.

Answer: C. Explanation: Ocean salinity rises with net evaporation and brine formation and falls with precipitation, river runoff, ice melt and mixing; it is a conservative tracer only within stated processes. The other options describe different processes, locations, scales or governance categories.

Q48. Which option avoids the main UPSC trap concerning Salinity budget?

A. During the southwest monsoon, strong winds and the Somali Current support coastal upwelling in the western Arabian Sea, linking circulation to nutrients and air-sea exchange. B. Positive and negative IOD phases shift thermocline, winds and convection differently, but neither phase guarantees a uniform Indian monsoon outcome because timing, ENSO and internal variability matter. C. The Indian Ocean Dipole is a coupled zonal contrast in sea-surface temperature and associated atmosphere-ocean conditions between western and eastern equatorial Indian Ocean sectors. D. Ocean salinity rises with net evaporation and brine formation and falls with precipitation, river runoff, ice melt and mixing; it is a conservative tracer only within stated processes.

Answer: D. Explanation: Ocean salinity rises with net evaporation and brine formation and falls with precipitation, river runoff, ice melt and mixing; it is a conservative tracer only within stated processes. The other options describe different processes, locations, scales or governance categories.

Q49. Which statement correctly explains Halocline-density link?

A. A halocline is a strong vertical salinity gradient; together with temperature it contributes to stratification, mixing resistance and the depth of the mixed layer. B. During the southwest monsoon, strong winds and the Somali Current support coastal upwelling in the western Arabian Sea, linking circulation to nutrients and air-sea exchange. C. North Indian Ocean surface circulation reverses or reorganises seasonally with monsoon winds, unlike a single fixed subtropical-gyre arrow map. D. The Arabian Sea is generally more saline than the Bay of Bengal because evaporation is stronger and freshwater input lower, while the Bay receives heavy rain and large river discharge.

Answer: A. Explanation: A halocline is a strong vertical salinity gradient; together with temperature it contributes to stratification, mixing resistance and the depth of the mixed layer. The other options describe different processes, locations, scales or governance categories.

Q50. Which option is the safest spatial interpretation of Halocline-density link?

A. During the southwest monsoon, strong winds and the Somali Current support coastal upwelling in the western Arabian Sea, linking circulation to nutrients and air-sea exchange. B. A halocline is a strong vertical salinity gradient; together with temperature it contributes to stratification, mixing resistance and the depth of the mixed layer. C. North Indian Ocean surface circulation reverses or reorganises seasonally with monsoon winds, unlike a single fixed subtropical-gyre arrow map. D. The Indian Ocean Dipole is a coupled zonal contrast in sea-surface temperature and associated atmosphere-ocean conditions between western and eastern equatorial Indian Ocean sectors.

Answer: B. Explanation: A halocline is a strong vertical salinity gradient; together with temperature it contributes to stratification, mixing resistance and the depth of the mixed layer. The other options describe different processes, locations, scales or governance categories.

Q51. Which statement preserves the process boundary for Halocline-density link?

A. The Indian Ocean Dipole is a coupled zonal contrast in sea-surface temperature and associated atmosphere-ocean conditions between western and eastern equatorial Indian Ocean sectors. B. Positive and negative IOD phases shift thermocline, winds and convection differently, but neither phase guarantees a uniform Indian monsoon outcome because timing, ENSO and internal variability matter. C. A halocline is a strong vertical salinity gradient; together with temperature it contributes to stratification, mixing resistance and the depth of the mixed layer. D. During the southwest monsoon, strong winds and the Somali Current support coastal upwelling in the western Arabian Sea, linking circulation to nutrients and air-sea exchange.

Answer: C. Explanation: A halocline is a strong vertical salinity gradient; together with temperature it contributes to stratification, mixing resistance and the depth of the mixed layer. The other options describe different processes, locations, scales or governance categories.

Q52. Which option avoids the main UPSC trap concerning Halocline-density link?

A. The Indian Ocean Dipole is a coupled zonal contrast in sea-surface temperature and associated atmosphere-ocean conditions between western and eastern equatorial Indian Ocean sectors. B. INCOIS describes OMNI and RAMA moorings that collect upper-ocean temperature, salinity, currents and meteorological variables for Indian Ocean monitoring and model validation. C. Positive and negative IOD phases shift thermocline, winds and convection differently, but neither phase guarantees a uniform Indian monsoon outcome because timing, ENSO and internal variability matter. D. A halocline is a strong vertical salinity gradient; together with temperature it contributes to stratification, mixing resistance and the depth of the mixed layer.

Answer: D. Explanation: A halocline is a strong vertical salinity gradient; together with temperature it contributes to stratification, mixing resistance and the depth of the mixed layer. The other options describe different processes, locations, scales or governance categories.

Q53. Which statement correctly explains Arabian Sea-Bay contrast?

A. The Arabian Sea is generally more saline than the Bay of Bengal because evaporation is stronger and freshwater input lower, while the Bay receives heavy rain and large river discharge. B. The Indian Ocean Dipole is a coupled zonal contrast in sea-surface temperature and associated atmosphere-ocean conditions between western and eastern equatorial Indian Ocean sectors. C. North Indian Ocean surface circulation reverses or reorganises seasonally with monsoon winds, unlike a single fixed subtropical-gyre arrow map. D. During the southwest monsoon, strong winds and the Somali Current support coastal upwelling in the western Arabian Sea, linking circulation to nutrients and air-sea exchange.

Answer: A. Explanation: The Arabian Sea is generally more saline than the Bay of Bengal because evaporation is stronger and freshwater input lower, while the Bay receives heavy rain and large river discharge. The other options describe different processes, locations, scales or governance categories.

Q54. Which option is the safest spatial interpretation of Arabian Sea-Bay contrast?

A. The Indian Ocean Dipole is a coupled zonal contrast in sea-surface temperature and associated atmosphere-ocean conditions between western and eastern equatorial Indian Ocean sectors. B. The Arabian Sea is generally more saline than the Bay of Bengal because evaporation is stronger and freshwater input lower, while the Bay receives heavy rain and large river discharge. C. Positive and negative IOD phases shift thermocline, winds and convection differently, but neither phase guarantees a uniform Indian monsoon outcome because timing, ENSO and internal variability matter. D. During the southwest monsoon, strong winds and the Somali Current support coastal upwelling in the western Arabian Sea, linking circulation to nutrients and air-sea exchange.

Answer: B. Explanation: The Arabian Sea is generally more saline than the Bay of Bengal because evaporation is stronger and freshwater input lower, while the Bay receives heavy rain and large river discharge. The other options describe different processes, locations, scales or governance categories.

Q55. Which statement preserves the process boundary for Arabian Sea-Bay contrast?

A. The Indian Ocean Dipole is a coupled zonal contrast in sea-surface temperature and associated atmosphere-ocean conditions between western and eastern equatorial Indian Ocean sectors. B. Positive and negative IOD phases shift thermocline, winds and convection differently, but neither phase guarantees a uniform Indian monsoon outcome because timing, ENSO and internal variability matter. C. The Arabian Sea is generally more saline than the Bay of Bengal because evaporation is stronger and freshwater input lower, while the Bay receives heavy rain and large river discharge. D. INCOIS describes OMNI and RAMA moorings that collect upper-ocean temperature, salinity, currents and meteorological variables for Indian Ocean monitoring and model validation.

Answer: C. Explanation: The Arabian Sea is generally more saline than the Bay of Bengal because evaporation is stronger and freshwater input lower, while the Bay receives heavy rain and large river discharge. The other options describe different processes, locations, scales or governance categories.

Q56. Which option avoids the main UPSC trap concerning Arabian Sea-Bay contrast?

A. Direct routed Mains demands are 2019 on ocean currents versus water masses and marine life, and 2022 on forces influencing currents and their fishing role. B. INCOIS describes OMNI and RAMA moorings that collect upper-ocean temperature, salinity, currents and meteorological variables for Indian Ocean monitoring and model validation. C. Positive and negative IOD phases shift thermocline, winds and convection differently, but neither phase guarantees a uniform Indian monsoon outcome because timing, ENSO and internal variability matter. D. The Arabian Sea is generally more saline than the Bay of Bengal because evaporation is stronger and freshwater input lower, while the Bay receives heavy rain and large river discharge.

Answer: D. Explanation: The Arabian Sea is generally more saline than the Bay of Bengal because evaporation is stronger and freshwater input lower, while the Bay receives heavy rain and large river discharge. The other options describe different processes, locations, scales or governance categories.

Q57. Which statement correctly explains North Indian Ocean reversal?

A. North Indian Ocean surface circulation reverses or reorganises seasonally with monsoon winds, unlike a single fixed subtropical-gyre arrow map. B. Positive and negative IOD phases shift thermocline, winds and convection differently, but neither phase guarantees a uniform Indian monsoon outcome because timing, ENSO and internal variability matter. C. During the southwest monsoon, strong winds and the Somali Current support coastal upwelling in the western Arabian Sea, linking circulation to nutrients and air-sea exchange. D. The Indian Ocean Dipole is a coupled zonal contrast in sea-surface temperature and associated atmosphere-ocean conditions between western and eastern equatorial Indian Ocean sectors.

Answer: A. Explanation: North Indian Ocean surface circulation reverses or reorganises seasonally with monsoon winds, unlike a single fixed subtropical-gyre arrow map. The other options describe different processes, locations, scales or governance categories.

Q58. Which option is the safest spatial interpretation of North Indian Ocean reversal?

A. The Indian Ocean Dipole is a coupled zonal contrast in sea-surface temperature and associated atmosphere-ocean conditions between western and eastern equatorial Indian Ocean sectors. B. North Indian Ocean surface circulation reverses or reorganises seasonally with monsoon winds, unlike a single fixed subtropical-gyre arrow map. C. Positive and negative IOD phases shift thermocline, winds and convection differently, but neither phase guarantees a uniform Indian monsoon outcome because timing, ENSO and internal variability matter. D. INCOIS describes OMNI and RAMA moorings that collect upper-ocean temperature, salinity, currents and meteorological variables for Indian Ocean monitoring and model validation.

Answer: B. Explanation: North Indian Ocean surface circulation reverses or reorganises seasonally with monsoon winds, unlike a single fixed subtropical-gyre arrow map. The other options describe different processes, locations, scales or governance categories.

Q59. Which statement preserves the process boundary for North Indian Ocean reversal?

A. Direct routed Mains demands are 2019 on ocean currents versus water masses and marine life, and 2022 on forces influencing currents and their fishing role. B. Positive and negative IOD phases shift thermocline, winds and convection differently, but neither phase guarantees a uniform Indian monsoon outcome because timing, ENSO and internal variability matter. C. North Indian Ocean surface circulation reverses or reorganises seasonally with monsoon winds, unlike a single fixed subtropical-gyre arrow map. D. INCOIS describes OMNI and RAMA moorings that collect upper-ocean temperature, salinity, currents and meteorological variables for Indian Ocean monitoring and model validation.

Answer: C. Explanation: North Indian Ocean surface circulation reverses or reorganises seasonally with monsoon winds, unlike a single fixed subtropical-gyre arrow map. The other options describe different processes, locations, scales or governance categories.

Q60. Which option avoids the main UPSC trap concerning North Indian Ocean reversal?

A. Direct routed Mains demands are 2019 on ocean currents versus water masses and marine life, and 2022 on forces influencing currents and their fishing role. B. INCOIS describes OMNI and RAMA moorings that collect upper-ocean temperature, salinity, currents and meteorological variables for Indian Ocean monitoring and model validation. C. An ocean current is a persistent, directed movement of seawater at a stated depth and time scale; a surface current, tidal stream, eddy and deep overturning branch are not interchangeable. D. North Indian Ocean surface circulation reverses or reorganises seasonally with monsoon winds, unlike a single fixed subtropical-gyre arrow map.

Answer: D. Explanation: North Indian Ocean surface circulation reverses or reorganises seasonally with monsoon winds, unlike a single fixed subtropical-gyre arrow map. The other options describe different processes, locations, scales or governance categories.

Q61. Which statement correctly explains Somali-current upwelling?

A. During the southwest monsoon, strong winds and the Somali Current support coastal upwelling in the western Arabian Sea, linking circulation to nutrients and air-sea exchange. B. Positive and negative IOD phases shift thermocline, winds and convection differently, but neither phase guarantees a uniform Indian monsoon outcome because timing, ENSO and internal variability matter. C. The Indian Ocean Dipole is a coupled zonal contrast in sea-surface temperature and associated atmosphere-ocean conditions between western and eastern equatorial Indian Ocean sectors. D. INCOIS describes OMNI and RAMA moorings that collect upper-ocean temperature, salinity, currents and meteorological variables for Indian Ocean monitoring and model validation.

Answer: A. Explanation: During the southwest monsoon, strong winds and the Somali Current support coastal upwelling in the western Arabian Sea, linking circulation to nutrients and air-sea exchange. The other options describe different processes, locations, scales or governance categories.

Q62. Which option is the safest spatial interpretation of Somali-current upwelling?

A. Direct routed Mains demands are 2019 on ocean currents versus water masses and marine life, and 2022 on forces influencing currents and their fishing role. B. During the southwest monsoon, strong winds and the Somali Current support coastal upwelling in the western Arabian Sea, linking circulation to nutrients and air-sea exchange. C. INCOIS describes OMNI and RAMA moorings that collect upper-ocean temperature, salinity, currents and meteorological variables for Indian Ocean monitoring and model validation. D. Positive and negative IOD phases shift thermocline, winds and convection differently, but neither phase guarantees a uniform Indian monsoon outcome because timing, ENSO and internal variability matter.

Answer: B. Explanation: During the southwest monsoon, strong winds and the Somali Current support coastal upwelling in the western Arabian Sea, linking circulation to nutrients and air-sea exchange. The other options describe different processes, locations, scales or governance categories.

Q63. Which statement preserves the process boundary for Somali-current upwelling?

A. Direct routed Mains demands are 2019 on ocean currents versus water masses and marine life, and 2022 on forces influencing currents and their fishing role. B. INCOIS describes OMNI and RAMA moorings that collect upper-ocean temperature, salinity, currents and meteorological variables for Indian Ocean monitoring and model validation. C. During the southwest monsoon, strong winds and the Somali Current support coastal upwelling in the western Arabian Sea, linking circulation to nutrients and air-sea exchange. D. An ocean current is a persistent, directed movement of seawater at a stated depth and time scale; a surface current, tidal stream, eddy and deep overturning branch are not interchangeable.

Answer: C. Explanation: During the southwest monsoon, strong winds and the Somali Current support coastal upwelling in the western Arabian Sea, linking circulation to nutrients and air-sea exchange. The other options describe different processes, locations, scales or governance categories.

Q64. Which option avoids the main UPSC trap concerning Somali-current upwelling?

A. Direct routed Mains demands are 2019 on ocean currents versus water masses and marine life, and 2022 on forces influencing currents and their fishing role. B. An ocean current is a persistent, directed movement of seawater at a stated depth and time scale; a surface current, tidal stream, eddy and deep overturning branch are not interchangeable. C. Planetary winds provide major surface stress, while Coriolis, pressure gradients, basin shape, friction and sea-level slope organise the resulting circulation. D. During the southwest monsoon, strong winds and the Somali Current support coastal upwelling in the western Arabian Sea, linking circulation to nutrients and air-sea exchange.

Answer: D. Explanation: During the southwest monsoon, strong winds and the Somali Current support coastal upwelling in the western Arabian Sea, linking circulation to nutrients and air-sea exchange. The other options describe different processes, locations, scales or governance categories.

Q65. Which statement correctly explains IOD definition?

A. The Indian Ocean Dipole is a coupled zonal contrast in sea-surface temperature and associated atmosphere-ocean conditions between western and eastern equatorial Indian Ocean sectors. B. Direct routed Mains demands are 2019 on ocean currents versus water masses and marine life, and 2022 on forces influencing currents and their fishing role. C. Positive and negative IOD phases shift thermocline, winds and convection differently, but neither phase guarantees a uniform Indian monsoon outcome because timing, ENSO and internal variability matter. D. INCOIS describes OMNI and RAMA moorings that collect upper-ocean temperature, salinity, currents and meteorological variables for Indian Ocean monitoring and model validation.

Answer: A. Explanation: The Indian Ocean Dipole is a coupled zonal contrast in sea-surface temperature and associated atmosphere-ocean conditions between western and eastern equatorial Indian Ocean sectors. The other options describe different processes, locations, scales or governance categories.

Q66. Which option is the safest spatial interpretation of IOD definition?

A. Direct routed Mains demands are 2019 on ocean currents versus water masses and marine life, and 2022 on forces influencing currents and their fishing role. B. The Indian Ocean Dipole is a coupled zonal contrast in sea-surface temperature and associated atmosphere-ocean conditions between western and eastern equatorial Indian Ocean sectors. C. An ocean current is a persistent, directed movement of seawater at a stated depth and time scale; a surface current, tidal stream, eddy and deep overturning branch are not interchangeable. D. INCOIS describes OMNI and RAMA moorings that collect upper-ocean temperature, salinity, currents and meteorological variables for Indian Ocean monitoring and model validation.

Answer: B. Explanation: The Indian Ocean Dipole is a coupled zonal contrast in sea-surface temperature and associated atmosphere-ocean conditions between western and eastern equatorial Indian Ocean sectors. The other options describe different processes, locations, scales or governance categories.

Q67. Which statement preserves the process boundary for IOD definition?

A. Planetary winds provide major surface stress, while Coriolis, pressure gradients, basin shape, friction and sea-level slope organise the resulting circulation. B. Direct routed Mains demands are 2019 on ocean currents versus water masses and marine life, and 2022 on forces influencing currents and their fishing role. C. The Indian Ocean Dipole is a coupled zonal contrast in sea-surface temperature and associated atmosphere-ocean conditions between western and eastern equatorial Indian Ocean sectors. D. An ocean current is a persistent, directed movement of seawater at a stated depth and time scale; a surface current, tidal stream, eddy and deep overturning branch are not interchangeable.

Answer: C. Explanation: The Indian Ocean Dipole is a coupled zonal contrast in sea-surface temperature and associated atmosphere-ocean conditions between western and eastern equatorial Indian Ocean sectors. The other options describe different processes, locations, scales or governance categories.

Q68. Which option avoids the main UPSC trap concerning IOD definition?

A. Planetary winds provide major surface stress, while Coriolis, pressure gradients, basin shape, friction and sea-level slope organise the resulting circulation. B. An ocean current is a persistent, directed movement of seawater at a stated depth and time scale; a surface current, tidal stream, eddy and deep overturning branch are not interchangeable. C. Temperature and salinity alter seawater density; cooling, evaporation, sea-ice processes, mixing and freshwater input help create sinking, spreading and overturning at depth. D. The Indian Ocean Dipole is a coupled zonal contrast in sea-surface temperature and associated atmosphere-ocean conditions between western and eastern equatorial Indian Ocean sectors.

Answer: D. Explanation: The Indian Ocean Dipole is a coupled zonal contrast in sea-surface temperature and associated atmosphere-ocean conditions between western and eastern equatorial Indian Ocean sectors. The other options describe different processes, locations, scales or governance categories.

Q69. Which statement correctly explains IOD phase boundary?

A. Positive and negative IOD phases shift thermocline, winds and convection differently, but neither phase guarantees a uniform Indian monsoon outcome because timing, ENSO and internal variability matter. B. Direct routed Mains demands are 2019 on ocean currents versus water masses and marine life, and 2022 on forces influencing currents and their fishing role. C. An ocean current is a persistent, directed movement of seawater at a stated depth and time scale; a surface current, tidal stream, eddy and deep overturning branch are not interchangeable. D. INCOIS describes OMNI and RAMA moorings that collect upper-ocean temperature, salinity, currents and meteorological variables for Indian Ocean monitoring and model validation.

Answer: A. Explanation: Positive and negative IOD phases shift thermocline, winds and convection differently, but neither phase guarantees a uniform Indian monsoon outcome because timing, ENSO and internal variability matter. The other options describe different processes, locations, scales or governance categories.

Q70. Which option is the safest spatial interpretation of IOD phase boundary?

A. Direct routed Mains demands are 2019 on ocean currents versus water masses and marine life, and 2022 on forces influencing currents and their fishing role. B. Positive and negative IOD phases shift thermocline, winds and convection differently, but neither phase guarantees a uniform Indian monsoon outcome because timing, ENSO and internal variability matter. C. An ocean current is a persistent, directed movement of seawater at a stated depth and time scale; a surface current, tidal stream, eddy and deep overturning branch are not interchangeable. D. Planetary winds provide major surface stress, while Coriolis, pressure gradients, basin shape, friction and sea-level slope organise the resulting circulation.

Answer: B. Explanation: Positive and negative IOD phases shift thermocline, winds and convection differently, but neither phase guarantees a uniform Indian monsoon outcome because timing, ENSO and internal variability matter. The other options describe different processes, locations, scales or governance categories.

Q71. Which statement preserves the process boundary for IOD phase boundary?

A. Temperature and salinity alter seawater density; cooling, evaporation, sea-ice processes, mixing and freshwater input help create sinking, spreading and overturning at depth. B. An ocean current is a persistent, directed movement of seawater at a stated depth and time scale; a surface current, tidal stream, eddy and deep overturning branch are not interchangeable. C. Positive and negative IOD phases shift thermocline, winds and convection differently, but neither phase guarantees a uniform Indian monsoon outcome because timing, ENSO and internal variability matter. D. Planetary winds provide major surface stress, while Coriolis, pressure gradients, basin shape, friction and sea-level slope organise the resulting circulation.

Answer: C. Explanation: Positive and negative IOD phases shift thermocline, winds and convection differently, but neither phase guarantees a uniform Indian monsoon outcome because timing, ENSO and internal variability matter. The other options describe different processes, locations, scales or governance categories.

Q72. Which option avoids the main UPSC trap concerning IOD phase boundary?

A. Warm and cold describe a current relative to surrounding water and source region, not an absolute temperature; each current also changes air-sea heat and moisture exchange. B. Planetary winds provide major surface stress, while Coriolis, pressure gradients, basin shape, friction and sea-level slope organise the resulting circulation. C. Temperature and salinity alter seawater density; cooling, evaporation, sea-ice processes, mixing and freshwater input help create sinking, spreading and overturning at depth. D. Positive and negative IOD phases shift thermocline, winds and convection differently, but neither phase guarantees a uniform Indian monsoon outcome because timing, ENSO and internal variability matter.

Answer: D. Explanation: Positive and negative IOD phases shift thermocline, winds and convection differently, but neither phase guarantees a uniform Indian monsoon outcome because timing, ENSO and internal variability matter. The other options describe different processes, locations, scales or governance categories.

Q73. Which statement correctly explains Official observing network?

A. INCOIS describes OMNI and RAMA moorings that collect upper-ocean temperature, salinity, currents and meteorological variables for Indian Ocean monitoring and model validation. B. Planetary winds provide major surface stress, while Coriolis, pressure gradients, basin shape, friction and sea-level slope organise the resulting circulation. C. An ocean current is a persistent, directed movement of seawater at a stated depth and time scale; a surface current, tidal stream, eddy and deep overturning branch are not interchangeable. D. Direct routed Mains demands are 2019 on ocean currents versus water masses and marine life, and 2022 on forces influencing currents and their fishing role.

Answer: A. Explanation: INCOIS describes OMNI and RAMA moorings that collect upper-ocean temperature, salinity, currents and meteorological variables for Indian Ocean monitoring and model validation. The other options describe different processes, locations, scales or governance categories.

Q74. Which option is the safest spatial interpretation of Official observing network?

A. Temperature and salinity alter seawater density; cooling, evaporation, sea-ice processes, mixing and freshwater input help create sinking, spreading and overturning at depth. B. INCOIS describes OMNI and RAMA moorings that collect upper-ocean temperature, salinity, currents and meteorological variables for Indian Ocean monitoring and model validation. C. Planetary winds provide major surface stress, while Coriolis, pressure gradients, basin shape, friction and sea-level slope organise the resulting circulation. D. An ocean current is a persistent, directed movement of seawater at a stated depth and time scale; a surface current, tidal stream, eddy and deep overturning branch are not interchangeable.

Answer: B. Explanation: INCOIS describes OMNI and RAMA moorings that collect upper-ocean temperature, salinity, currents and meteorological variables for Indian Ocean monitoring and model validation. The other options describe different processes, locations, scales or governance categories.

Q75. Which statement preserves the process boundary for Official observing network?

A. Warm and cold describe a current relative to surrounding water and source region, not an absolute temperature; each current also changes air-sea heat and moisture exchange. B. Temperature and salinity alter seawater density; cooling, evaporation, sea-ice processes, mixing and freshwater input help create sinking, spreading and overturning at depth. C. INCOIS describes OMNI and RAMA moorings that collect upper-ocean temperature, salinity, currents and meteorological variables for Indian Ocean monitoring and model validation. D. Planetary winds provide major surface stress, while Coriolis, pressure gradients, basin shape, friction and sea-level slope organise the resulting circulation.

Answer: C. Explanation: INCOIS describes OMNI and RAMA moorings that collect upper-ocean temperature, salinity, currents and meteorological variables for Indian Ocean monitoring and model validation. The other options describe different processes, locations, scales or governance categories.

Q76. Which option avoids the main UPSC trap concerning Official observing network?

A. Wind stress and Coriolis organise subtropical gyres clockwise in the Northern Hemisphere and anticlockwise in the Southern Hemisphere, with equatorial and seasonal complications. B. Temperature and salinity alter seawater density; cooling, evaporation, sea-ice processes, mixing and freshwater input help create sinking, spreading and overturning at depth. C. Warm and cold describe a current relative to surrounding water and source region, not an absolute temperature; each current also changes air-sea heat and moisture exchange. D. INCOIS describes OMNI and RAMA moorings that collect upper-ocean temperature, salinity, currents and meteorological variables for Indian Ocean monitoring and model validation.

Answer: D. Explanation: INCOIS describes OMNI and RAMA moorings that collect upper-ocean temperature, salinity, currents and meteorological variables for Indian Ocean monitoring and model validation. The other options describe different processes, locations, scales or governance categories.

Q77. Which statement correctly explains Verified current PYQ routes?

A. Direct routed Mains demands are 2019 on ocean currents versus water masses and marine life, and 2022 on forces influencing currents and their fishing role. B. An ocean current is a persistent, directed movement of seawater at a stated depth and time scale; a surface current, tidal stream, eddy and deep overturning branch are not interchangeable. C. Temperature and salinity alter seawater density; cooling, evaporation, sea-ice processes, mixing and freshwater input help create sinking, spreading and overturning at depth. D. Planetary winds provide major surface stress, while Coriolis, pressure gradients, basin shape, friction and sea-level slope organise the resulting circulation.

Answer: A. Explanation: Direct routed Mains demands are 2019 on ocean currents versus water masses and marine life, and 2022 on forces influencing currents and their fishing role. The other options describe different processes, locations, scales or governance categories.

Q78. Which option is the safest spatial interpretation of Verified current PYQ routes?

A. Warm and cold describe a current relative to surrounding water and source region, not an absolute temperature; each current also changes air-sea heat and moisture exchange. B. Direct routed Mains demands are 2019 on ocean currents versus water masses and marine life, and 2022 on forces influencing currents and their fishing role. C. Planetary winds provide major surface stress, while Coriolis, pressure gradients, basin shape, friction and sea-level slope organise the resulting circulation. D. Temperature and salinity alter seawater density; cooling, evaporation, sea-ice processes, mixing and freshwater input help create sinking, spreading and overturning at depth.

Answer: B. Explanation: Direct routed Mains demands are 2019 on ocean currents versus water masses and marine life, and 2022 on forces influencing currents and their fishing role. The other options describe different processes, locations, scales or governance categories.

Q79. Which statement preserves the process boundary for Verified current PYQ routes?

A. Temperature and salinity alter seawater density; cooling, evaporation, sea-ice processes, mixing and freshwater input help create sinking, spreading and overturning at depth. B. Wind stress and Coriolis organise subtropical gyres clockwise in the Northern Hemisphere and anticlockwise in the Southern Hemisphere, with equatorial and seasonal complications. C. Direct routed Mains demands are 2019 on ocean currents versus water masses and marine life, and 2022 on forces influencing currents and their fishing role. D. Warm and cold describe a current relative to surrounding water and source region, not an absolute temperature; each current also changes air-sea heat and moisture exchange.

Answer: C. Explanation: Direct routed Mains demands are 2019 on ocean currents versus water masses and marine life, and 2022 on forces influencing currents and their fishing role. The other options describe different processes, locations, scales or governance categories.

Q80. Which option avoids the main UPSC trap concerning Verified current PYQ routes?

A. Warm and cold describe a current relative to surrounding water and source region, not an absolute temperature; each current also changes air-sea heat and moisture exchange. B. Wind stress and Coriolis organise subtropical gyres clockwise in the Northern Hemisphere and anticlockwise in the Southern Hemisphere, with equatorial and seasonal complications. C. Western boundary currents such as the Gulf Stream and Kuroshio are generally narrow, swift and deep because basin-scale vorticity balance concentrates return flow on the west. D. Direct routed Mains demands are 2019 on ocean currents versus water masses and marine life, and 2022 on forces influencing currents and their fishing role.

Answer: D. Explanation: Direct routed Mains demands are 2019 on ocean currents versus water masses and marine life, and 2022 on forces influencing currents and their fishing role. The other options describe different processes, locations, scales or governance categories.

PYQS AND ANSWER PRACTICE

VERIFIED PYQ OWNERSHIP AUDIT

Verified direct routes retain the 2019 GS-I ocean-current and water-mass demand and the 2022 GS-I current-forces and fishing demand. No current IOD phase, forecast value or official objective answer letter is inferred.

OWNER PYQ LEDGER EXTRACTS

Dead Zones & Marine Hypoxia (PYQ gap - beyond core books)

ANALYSIS: **Source note:** Dead zones / hypoxia / oxygen-minimum zones are **NOT** in Leong, Husain or Khullar. The books cover only the **precursor, eutrophication** (Husain: "a process in which lakes receive nutrients and become enriched"; Khullar: nitrate/phosphate runoff -> "eutrophication"). The dead-zone mechanism below is standard oceanography (ANALYSIS:) + current data (CURRENT:), added to close a UPSC GS-1 gap.

- ANALYSIS: **Dead zone** = a stretch of ocean/lake with **very low dissolved oxygen (hypoxia, DO < ~2 mg/L)** where most marine animals cannot survive; severe cases are **anoxic** (no O₂).
- FACT:->ANALYSIS: **Mechanism chain:** nutrient runoff (**nitrogen + phosphorus** from fertilizer/sewage) -> **eutrophication** (FACT: book concept) -> **algal bloom** -> algae die -> **bacterial decomposition consumes oxygen** -> **hypoxia** -> **dead zone**. Made worse by **stratification** (warm light surface water caps colder O₂-poor deep water) and **ocean warming** (warm water holds less O₂).

Type	Cause	Depth	Example
ANALYSIS: Coastal / anthropogenic dead zone	Eutrophication from land runoff	Shallow, seasonal	CURRENT: Gulf of Mexico (Mississippi fertilizer runoff), Baltic Sea
ANALYSIS: Oxygen Minimum Zone (OMZ)	High productivity, respiration and weak ventilation; human forcing can intensify it	Mid-depth ocean	CURRENT: Arabian Sea (one of the strongest tropical OMZs - see advanced file)

MEMORY: Trap: Hypoxia means too little oxygen. Do not force a binary rule that every OMZ is "natural" and every coastal dead zone "human-made"; circulation, productivity, warming and nutrient loading can interact.

Ocean-floor Relief: Seamount & Guyot (PYQ gap)

FACT: Grounded in GC Leong + Majid Husain + ANALYSIS: standard geography. Added to close a UPSC gap.

- FACT: GC Leong states ocean basins have submarine ridges, plateaux, canyons, plains and trenches, broadly comparable to land relief.
- FACT: Majid Husain lists major ocean-floor features: continental shelf, slope, rise, abyssal floor, trenches, ridges and **seamounts**.
- ANALYSIS: **Seamount** = isolated submarine volcanic mountain rising from the ocean floor but not reaching sea level.
- ANALYSIS: **Guyot** = flat-topped seamount, usually an eroded former volcanic island later subsided below sea level.

Feature	Shape	Formation clue	Exam trap
FACT:/ANALYSIS: Seamount	Conical submarine high	Volcanic build-up	Not an island unless it emerges
ANALYSIS: Guyot	Flat-topped submarine high	Wave erosion + subsidence	Not a trench or ridge
FACT: Trench	Long narrow deep	Subduction zone	Deepest ocean-floor feature

MEMORY: Trap: A guyot's flat top suggests it was once near/above sea level; it is not flat because sediment simply filled a basin.

Recent PYQ Integration (2024-2025)

Status: 2024-2025 question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-PRELIMS-2024-2025.md. **Answer-key rule:** The official 2024-2025 Prelims Set-A keys are present in the repository and CSAT Set-A keys are supplied; even so, no option or answer is recorded or inferred in this integration.

- **Years represented:** 2024
- **Paper(s):** Prelims GS-I
- **Routed question demands:** 1

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2024	Prelims GS-I	89	Red Sea - low precipitation and absence of major river inflow	Objective question; official Set-A key available locally, answer not inferred	Key available locally (official Set-A answer key present); answer not recorded here	Cover the named fact/concept and its likely statement-level distinctions.

What this owner must now support

- Red Sea - low precipitation and absence of major river inflow

This block integrates the 2024-2025 examinable demand and paper metadata. It is kept separate from the 2018-2023 block and does not convert an unkeyed/answer-free objective question into a solved answer.

Historical PYQ Integration (2018-2023)

Status: Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-MAINS-GS1-GS2-ESSAY-2018-2023.md, _PYQ-ROUTING-PRELIMS-2018-2023.md. **Answer-key rule:** The official 2018-2023 Prelims/CSAT keys are not held locally; no option or answer has been inferred.

- **Years represented:** 2019, 2020, 2021, 2022
- **Paper(s):** GS-I, Prelims GS-I
- **Routed question demands:** 6

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2019	GS-I	17	Ocean currents and water masses and their impact on marine life	How do they differ with examples · 15 marks · 250 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2020	Prelims GS-I	93	Ocean Mean Temperature Indian Ocean monsoon prediction tool	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2021	Prelims GS-I	56	Ocean seabed mineral exploration licensing international waters	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2021	Prelims GS-I	58	Ocean surface temperature patterns trade winds westerlies	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2021	Prelims GS-I	62	Global water distribution rivers groundwater polar ice	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2022	GS-I	14	Forces influencing ocean currents and their role in fishing	What are and Describe · 15 marks · 250 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.

What this owner must now support

- Ocean currents and water masses and their impact on marine life
- Ocean Mean Temperature Indian Ocean monsoon prediction tool
- Ocean seabed mineral exploration licensing international waters
- Ocean surface temperature patterns trade winds westerlies
- Global water distribution rivers groundwater polar ice
- Forces influencing ocean currents and their role in fishing

The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

□ The Arabian Sea Dead Zone / OMZ (PYQ gap - beyond core books)

ANALYSIS: **Source note:** Not covered in Khullar/Husain (books stop at *eutrophication*). Added to close a UPSC GS-1 gap; ANALYSIS: = standard oceanography, CURRENT: = current data.

- ANALYSIS: The **Arabian Sea** hosts one of the world's strongest tropical oxygen-minimum zones at intermediate depth. Avoid a single area estimate for a moving three-dimensional low-oxygen volume.
- ANALYSIS: **Why here:** monsoon-driven productivity and organic-matter decay combine with weak ventilation. Warming, circulation change and nutrient inputs may intensify coastal or upper-ocean deoxygenation.
- ANALYSIS: **Consequences:** loss of fixed nitrogen via **denitrification/anammox**, release of the greenhouse gas **nitrous oxide (N₂O)**, fishery shifts, and replacement of diatoms by hypoxia-tolerant **Noctiluca (green Noctiluca blooms)** - an India-relevant fisheries/climate concern.

Feature	Coastal dead zone	Arabian Sea OMZ
ANALYSIS: Driver	Human eutrophication (land runoff)	Natural upwelling + weak ventilation
ANALYSIS: Location	Shallow shelf, seasonal	Mid-depth, near-permanent
ANALYSIS: India link	Estuaries/coasts near cities	Fisheries, monsoon-productivity, N ₂ O emissions

MEMORY: Trap: The Arabian Sea OMZ has a strong natural oceanographic basis, but human nutrient inputs and climate change can modify it; do not present “natural” and “anthropogenic” as mutually exclusive.

PYQ DEMAND CARD 1 - 2019 GS-I

Demand: How do ocean currents and water masses differ in their impacts on marine life and the coastal environment? Give suitable examples. (250 words)

Status: Verified routed direct Mains demand.

Model solution: Define currents as moving flows and water masses as bodies with characteristic temperature-salinity properties. Compare heat transport, fronts, upwelling, oxygen and nutrient pathways; use Gulf Stream, Peru upwelling and Arabian Sea-Bay contrasts. Qualify that ecology also depends on light, habitat and fishing pressure.

PYQ DEMAND CARD 2 - 2022 GS-I

Demand: What are the forces that influence ocean currents? Describe their role in the fishing industry of the world. (250 words)

Status: Verified routed direct Mains demand.

Model solution: Group wind stress, Coriolis, pressure gradient, density, gravity, friction and basin geometry. Link upwelling and fronts to nutrient supply and fish aggregation, using Peru, Benguela, Somali and Kuroshio-Oyashio examples. Qualify with oxygen, seasonality, habitat and management.

ORIGINAL MAINS 1 - 10 MARKS

Question: Explain why surface currents cannot be derived from wind direction alone. Answer in about 150 words.

Model thesis: Wind stress initiates motion, but Coriolis, pressure gradients, friction and basin geometry turn it into organised currents and gyres.

Claim -> named evidence -> analysis -> qualification:

- Planetary winds provide major surface stress, while Coriolis, pressure gradients, basin shape, friction and sea-level slope organise the resulting circulation.
- Wind stress and Coriolis organise subtropical gyres clockwise in the Northern Hemisphere and anticlockwise in the Southern Hemisphere, with equatorial and seasonal complications.

Qualified conclusion: Wind stress initiates motion, but Coriolis, pressure gradients, friction and basin geometry turn it into organised currents and gyres.

ORIGINAL MAINS 2 - 10 MARKS

Question: Differentiate spring and neap tides and explain why local tidal range still varies. Answer in about 150 words.

Model thesis: Astronomical alignment sets the cycle, while shelf-basin geometry, resonance and friction determine local expression.

Claim -> named evidence -> analysis -> qualification:

- Tides arise chiefly from the differential gravitational effects of the Moon and Sun together with Earth-Moon system dynamics, then are reshaped by basin geometry and friction.
- Spring tides occur near new and full moon when solar and lunar tidal effects reinforce; neap tides occur near quarter moons when the effects partly offset.
- Tidal range depends on astronomical forcing, coastline and basin resonance, shelf depth, friction and local timing; funnel-shaped bays can amplify range.

Qualified conclusion: Astronomical alignment sets the cycle, while shelf-basin geometry, resonance and friction determine local expression.

ORIGINAL MAINS 3 - 15 MARKS

Question: Analyse how temperature and salinity create ocean stratification and deep circulation. Answer in about 250 words.

Model thesis: Heat and freshwater budgets alter density, producing mixed layers, haloclines and sinking-spreading pathways modified by mixing and topography.

Claim -> named evidence -> analysis -> qualification:

- Temperature and salinity alter seawater density; cooling, evaporation, sea-ice processes, mixing and freshwater input help create sinking, spreading and overturning at depth.
- Ocean salinity rises with net evaporation and brine formation and falls with precipitation, river runoff, ice melt and mixing; it is a conservative tracer only within stated processes.
- A halocline is a strong vertical salinity gradient; together with temperature it contributes to stratification, mixing resistance and the depth of the mixed layer.

Qualified conclusion: Heat and freshwater budgets alter density, producing mixed layers, haloclines and sinking-spreading pathways modified by mixing and topography.

ORIGINAL MAINS 4 - 15 MARKS

Question: Explain the Arabian Sea-Bay of Bengal contrast in salinity and circulation. Answer in about 250 words.

Model thesis: Evaporation, rainfall, river runoff and monsoon winds generate contrasting stratification and seasonal current responses across the two basins.

Claim -> named evidence -> analysis -> qualification:

- The Arabian Sea is generally more saline than the Bay of Bengal because evaporation is stronger and freshwater input lower, while the Bay receives heavy rain and large river discharge.
- North Indian Ocean surface circulation reverses or reorganises seasonally with monsoon winds, unlike a single fixed subtropical-gyre arrow map.
- During the southwest monsoon, strong winds and the Somali Current support coastal upwelling in the western Arabian Sea, linking circulation to nutrients and air-sea exchange.

Qualified conclusion: Evaporation, rainfall, river runoff and monsoon winds generate contrasting stratification and seasonal current responses across the two basins.

ORIGINAL MAINS 5 - 20 MARKS

Question: Examine the influence of ocean currents on climate, upwelling and fisheries. Answer in about 300 words.

Model thesis: Currents redistribute heat and nutrients, but marine productivity depends on light, mixing, oxygen, habitat and fishing pressure as well as circulation.

Claim -> named evidence -> analysis -> qualification:

- Warm and cold describe a current relative to surrounding water and source region, not an absolute temperature; each current also changes air-sea heat and moisture exchange.
- Western boundary currents such as the Gulf Stream and Kuroshio are generally narrow, swift and deep because basin-scale vorticity balance concentrates return flow on the west.
- Eastern boundary currents are generally broad and cool, and alongshore winds can drive Ekman transport away from the coast so nutrient-rich deeper water rises.
- Productive fisheries often occur in upwelling zones or mixing fronts because nutrients support food webs; current convergence alone does not guarantee a rich fishery.

Qualified conclusion: Currents redistribute heat and nutrients, but marine productivity depends on light, mixing, oxygen, habitat and fishing pressure as well as circulation.

ORIGINAL MAINS 6 - 20 MARKS

Question: Assess the Indian Ocean Dipole as a modifier of the Indian monsoon rather than a deterministic switch. Answer in about 300 words.

Model thesis: IOD couples SST, thermocline, winds and convection, yet impacts depend on phase timing, amplitude, ENSO-MJO interaction and regional circulation.

Claim -> named evidence -> analysis -> qualification:

- The Indian Ocean Dipole is a coupled zonal contrast in sea-surface temperature and associated atmosphere-ocean conditions between western and eastern equatorial Indian Ocean sectors.
- Positive and negative IOD phases shift thermocline, winds and convection differently, but neither phase guarantees a uniform Indian monsoon outcome because timing, ENSO and internal variability matter.
- INCOIS describes OMNI and RAMA moorings that collect upper-ocean temperature, salinity, currents and meteorological variables for Indian Ocean monitoring and model validation.
- Direct routed Mains demands are 2019 on ocean currents versus water masses and marine life, and 2022 on forces influencing currents and their fishing role.

Qualified conclusion: IOD couples SST, thermocline, winds and convection, yet impacts depend on phase timing, amplitude, ENSO-MJO interaction and regional circulation.